

MICHAEL EMEFIENEM

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SOFTWARE ENGINEER

Software Engineer with 3 years of experience building production backend and full-stack systems with TypeScript, Node.js, PostgreSQL, and Redis. Experienced designing reliable APIs and distributed workflows involving payments, idempotency, concurrency, state management, third-party integrations, and failure recovery. Built and shipped systems end-to-end across backend architecture, infrastructure, testing, and product features, with additional experience integrating AI APIs and asynchronous workflows.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, SQL
- **Frontend:** React, Next.js, Vue.js, Tailwind CSS, ShadCN
- **Backend:** Node.js, Express.js, PostgreSQL, Prisma ORM, Drizzle ORM
- **AI:** OpenAI, Claude, Gemini, OpenRouter, AI API integration
- **Tools:** Git, Docker, AWS, Redux, React Query, RTK Query, Zustand, Jest, Vitest, React Testing Library

PROFESSIONAL EXPERIENCE

EBINUM – Software Engineer

Lagos, Nigeria | Jan 2026 – Present

- Designed and built EBINUM end-to-end, owning product architecture, backend services, payment workflows, transaction intelligence, reliability, and merchant infrastructure.
- Built payment flows for intents, confirmation, processing, refunds, disputes, and payouts, with Stripe and Adyen as configurable payment providers that merchants can connect and choose based on their preference.
- Implemented distributed locking, idempotent request handling, and multi-state transaction reconciliation across two external payment providers (Stripe, Adyen) with differing APIs and failure semantics — designing internal abstractions to normalize behavior across both.
- Built a microservice backend using TypeScript/Node.js, PostgreSQL, Prisma, and Redis, with transaction intelligence and failure-recovery capabilities.
- Designed failure recovery, auditability, and merchant onboarding/KYC workflows for a production-oriented payment system.
- Used test suites to define and validate expected system behavior during design, not just post-implementation, particularly around payment state transitions and failure recovery paths.

Axendit Limited – Software Engineer

Lagos, Nigeria | March 2024 – September 2026

- Built and maintained production features across frontend and backend systems using Next.js, TypeScript, React, Node.js, PostgreSQL, Prisma, and Redis.
- Developed a multi-role school management platform supporting 6 user types, building dashboards and workflows across 10+ modules with Redux and RTK Query.
- Implemented real-time functionality with WebSockets, authentication with JWT, and role-based access control for different user workflows.
- Built backend REST APIs for a document management system using Node.js, Express, PostgreSQL, and Prisma, including document lifecycle management, folder/versioning workflows, permissions, and audit logging.
- Added Redis caching to frequently accessed backend data and designed access-control mechanisms around sensitive documents and user roles.
- Contributed to a nationwide consumer protection portal for the NCAA, improving frontend workflows and usability for complaint-management processes.
- Worked across architecture, testing, debugging, and feature delivery, using Vitest and collaborating on technical decisions throughout the development lifecycle.

Bond – Frontend Developer (Contract)

Remote | January 2026 – June 2026

- Led frontend development for the company's marketing website and internal back-office platform using React and TypeScript.
- Designed reusable component systems and scalable frontend architecture that accelerated feature development.
- Built merchant dashboard functionality that improved transaction visibility and streamlined operational processes.
- Integrated frontend applications with backend APIs while maintaining clean state management and modular code organization.
- Delivered production-ready features under tight deadlines in a fast-paced startup environment.

Zidio Development – Web Developer Intern

Remote | June 2023 – February 2024

- Developed responsive web applications using React, JavaScript, and Redux, improving user experience across desktop and mobile devices.
- Collaborated with senior developers to deliver client-facing functionality and resolve production issues.

OPEN SOURCE CONTRIBUTIONS

- **OpenSRE (Python):** Fixed data validation for list-shaped payloads and updated the validation test suite while preserving compatibility across existing payload structures.
- **DevTools Hub (React, TypeScript):** Built a bidirectional JSON ↔ CSV converter with validation and integrated it into the project's existing tool catalog.
- **gmm-haproxy-vscode (TypeScript):** Expanded parser and validator test coverage by adding a production-style HAProxy TLS configuration fixture.

SELECTED PROJECTS

AI Workflow Automation Platform | Next.js, TypeScript, React Flow, tRPC, Prisma, Postgres, Inngest, Better Auth

- Built an AI-powered no-code workflow automation platform that enables users to visually design and execute complex business automations through a drag-and-drop interface inspired by Zapier and n8n.
- Designed a node-based workflow execution engine supporting webhook, scheduled, and manual triggers, with extensible integrations for external services and APIs.
- Integrated third-party AI APIs including OpenAI, Claude, and Gemini, building resilient abstractions to support content generation, summarization, and classification workflows.
- Implemented reliable background job processing using Inngest to execute asynchronous workflows and long-running tasks at scale.
- Developed subscription and usage control features, including authentication, access management, and payment integration for SaaS monetization.

AI Wireframe-to-Code Platform | Next.js, TypeScript, OpenRouter AI, PostgreSQL, Drizzle ORM, Tailwind CSS

- Built an AI-powered wireframe-to-code platform that converts uploaded UI wireframes into editable React and Tailwind CSS components.
- Integrated OpenRouter AI models to generate frontend code dynamically from uploaded wireframe inputs.

Live Demo: <https://wireframe-to-code-emefienem.vercel.app/>

EDUCATION

LAGOS STATE UNIVERSITY

Lagos, Nigeria | July 2021 – September 2025

Bachelor of Science (B.Sc.) Computer Science – CGPA: 4.36/5.0

